Information Management II

CSU 34041

9. Mapping to Logical Design

Overview

- Introduction to Relational Schema Mapping
- Mapping Entity Types
- Mapping Multivalued Attributes
- Mapping Relationships
- Cinema Example

Relational Schema Mapping

Relational Schema Mapping

- How to move from a conceptual database design
  - Entity Relationship **Model**
- …to a logical database design
  - Relational Database **Schema**
- We follow a series of steps to map entity types, relationships and attributes into relations

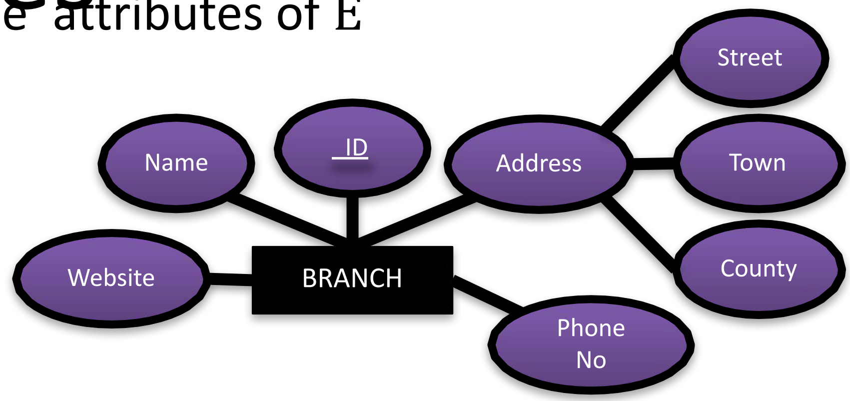
Relational Schema Mapping

- We will use the examples from the previous lectures to illustrate these mapping steps
- The mapping will create:
  - Relations
    - with simple, single-valued attributes
  - Constraints
    - primary keys
    - secondary unique keys
    - referential integrity constraints

Mapping Entity Types

Mapping of Entity Types

- For each entity type E in the ER diagram, create a relation R that includes all the simple attributes of E

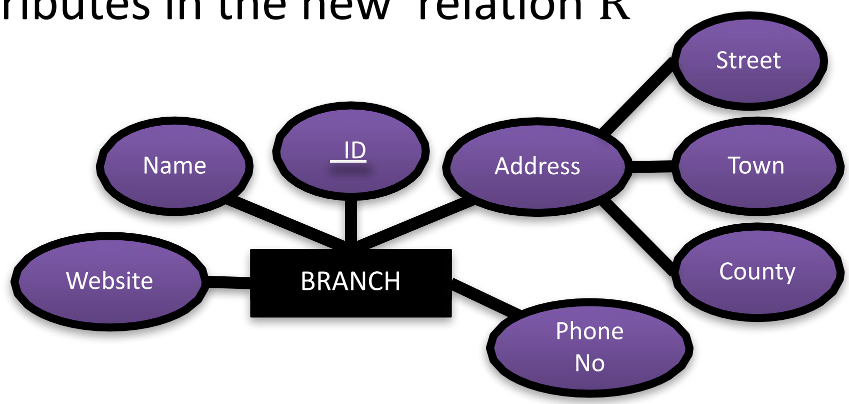


BRANCH

branch_id	name	street	town	county	phone_no	website

Mapping of Entity Types

- Composite attributes
  – when mapping composite attributes include only the simple component attributes in the new relation R



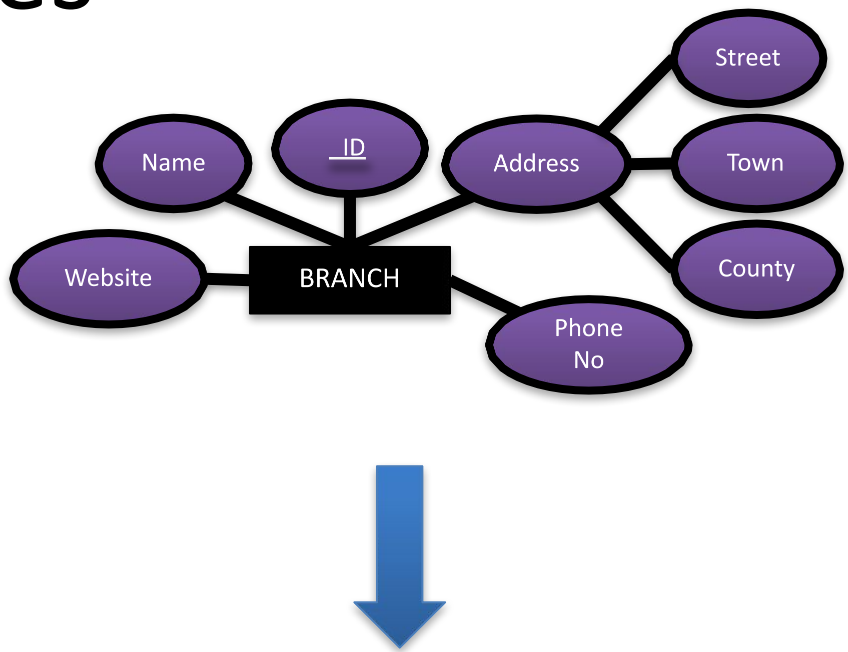
BRANCH

branch_id	name	street	town	county	phone_no	website

Mapping of Entity Types

- Key attributes
  - choose one of the key attributes of E as the primary key of R
  - composite key attributes can be included as a *composite primary key*
- Additional key attributes should be included as secondary unique keys of the relation

Mapping of Entity Types



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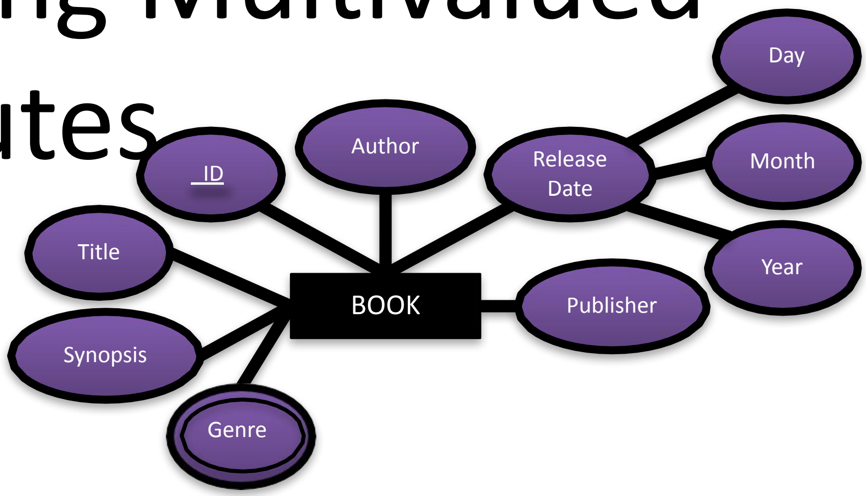
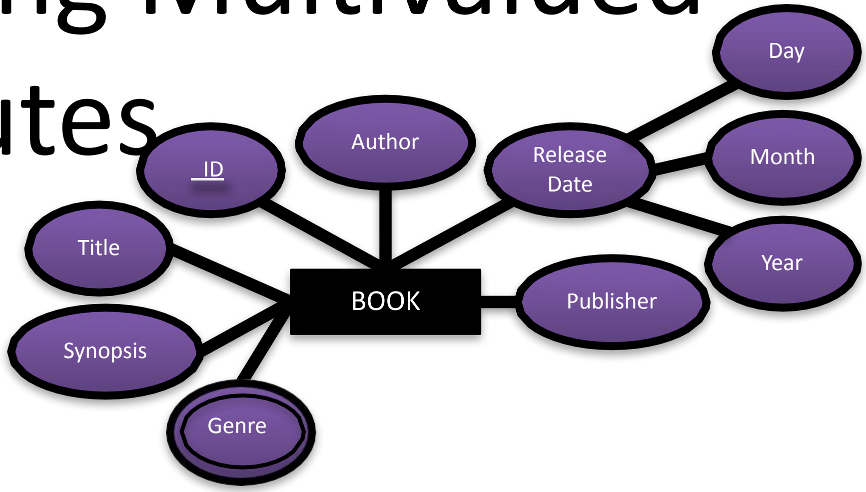
Mapping Multivalued Attributes

Mapping Multivalued Attributes



- For each multivalued attribute A, create a new relation R
- The new relation R will include:
  - An attribute corresponding to A
  - The primary key K from the relation that represents the entity type that A came from
    - This becomes a *foreign key* in R
  - The *primary key* of R is the combination of A and K

Mapping Multivalued Attributes



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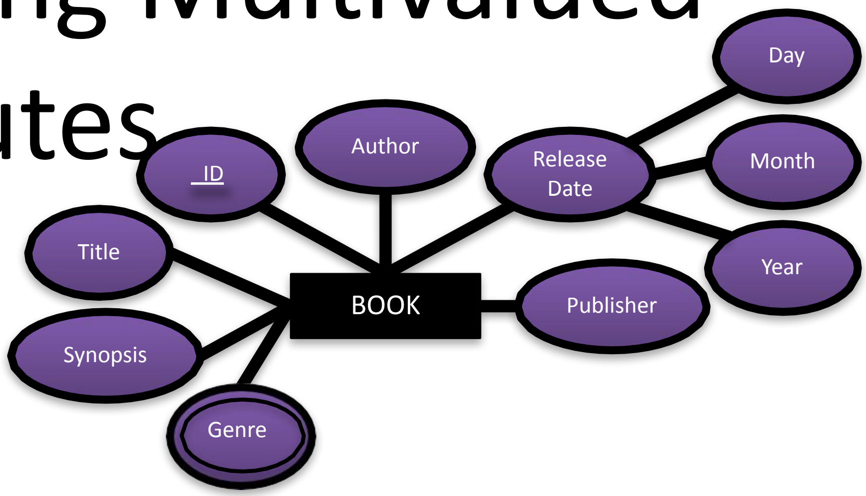
BOOK

book_id	title	synopsis	author	cert	publisher	release_day	release_month	release_year

Mapping Multivalued Attributes



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GENRE

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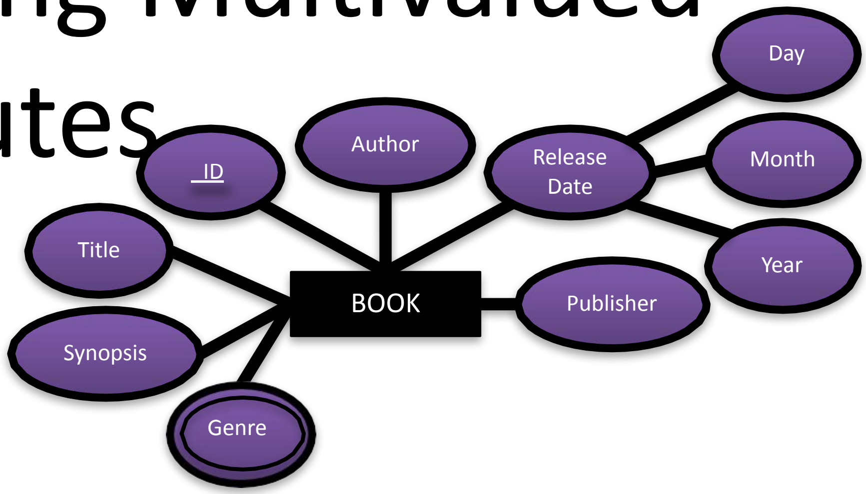

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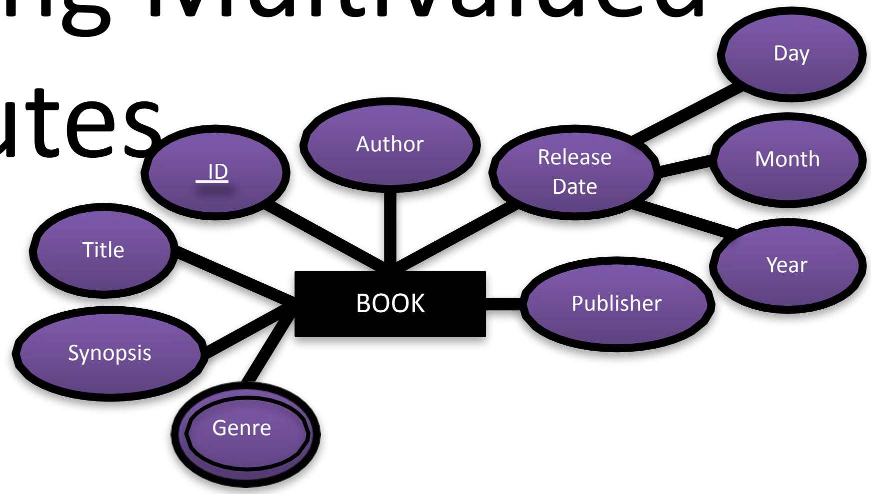

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genre

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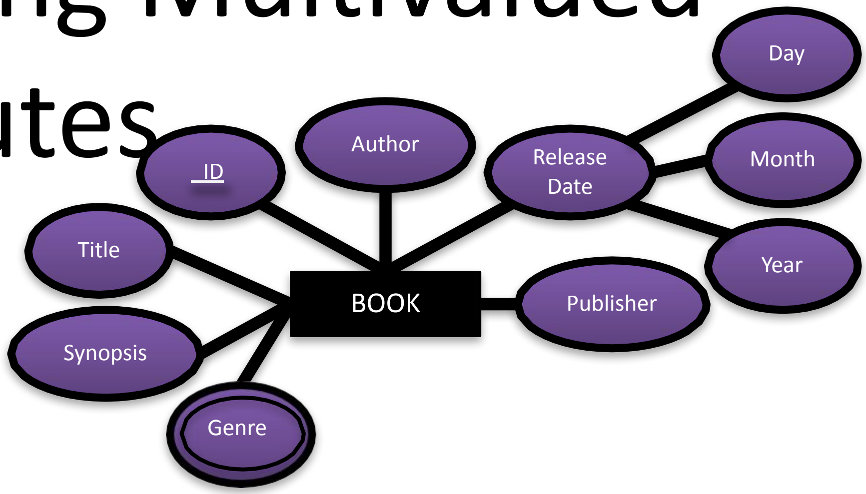

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genre

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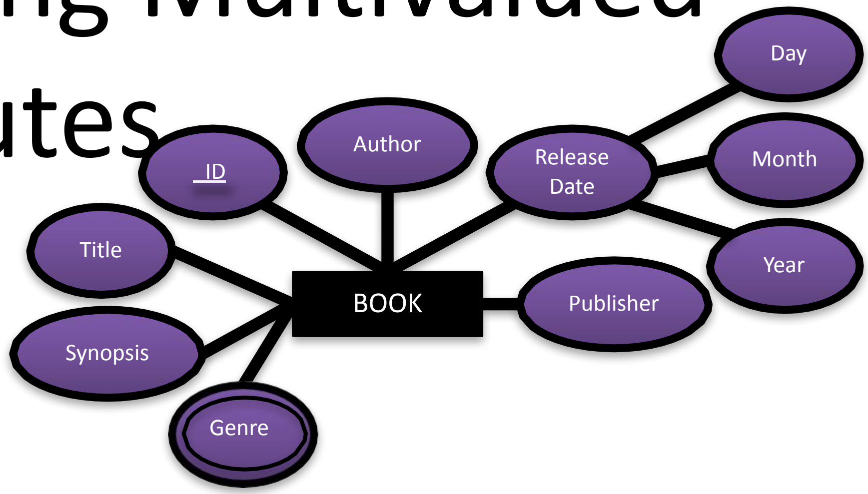

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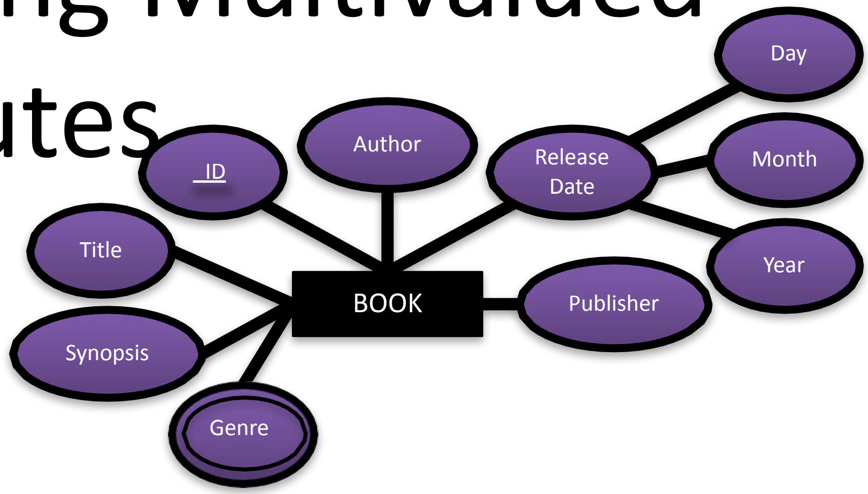

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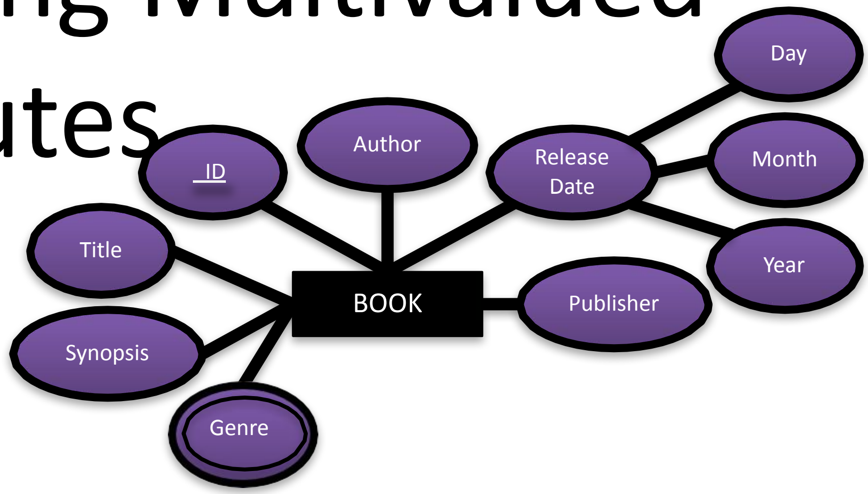
book_id	genre

19

Mapping Multivalued Attributes



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GENRE

book_id	genre

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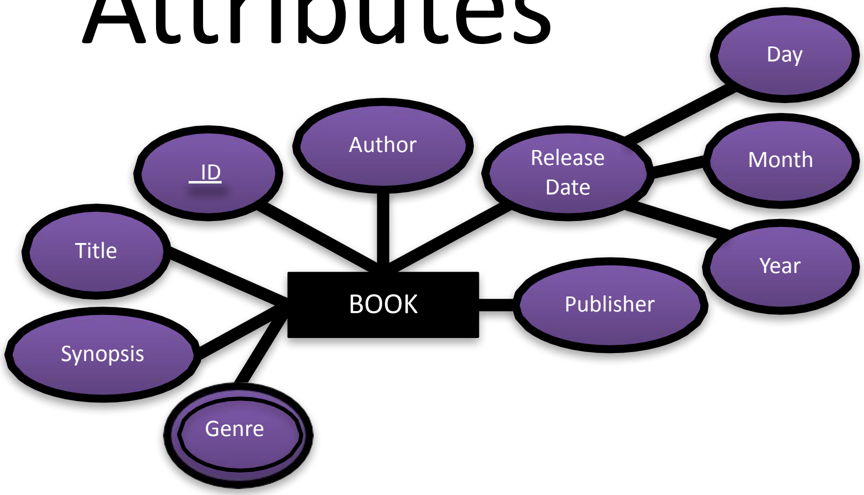

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book_id	genre

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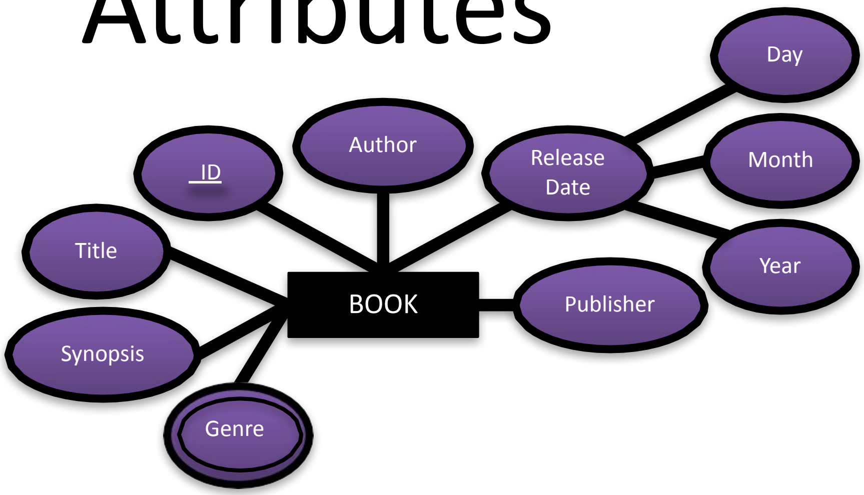
Mapping Multivalued Attributes



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GENRE

book_id	genre

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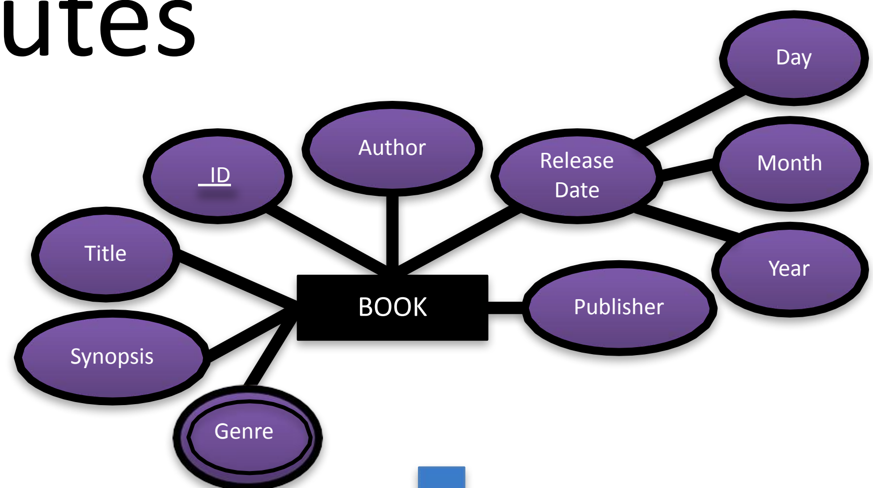

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GENRE

book_id	genre

Mapping Relationships

Mapping Relationships

- In addition to mapping the entity types from the ER model into the Relational Schema, we also need to map the relationship types

- Each relationship type is modeled differently

  - 1:1 One to One
  - 1:N One to Many
  - M:N Many to Many

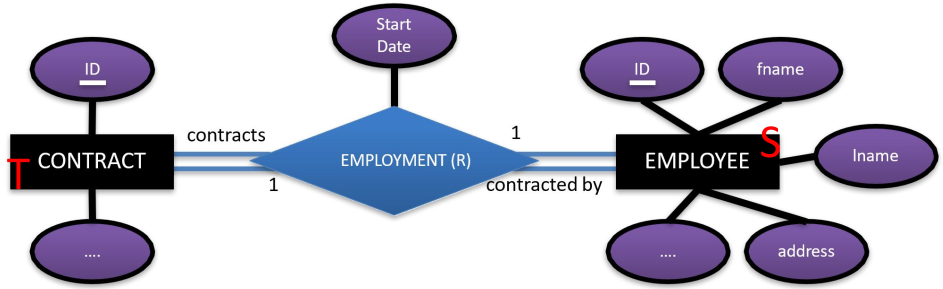
Mapping 1:1 Relationships

Mapping 1:1 Relationships

- There are two main approaches to mapping 1:1 relationships
  - Foreign Key Approach
    - Most commonly used
  - Merged-Relation Approach
    - Used in cases of *total participation*

Mapping 1:1 Relationships

- For each 1:1 relationship type R
  - identify the relations S and T that correspond to the entity types participating in R



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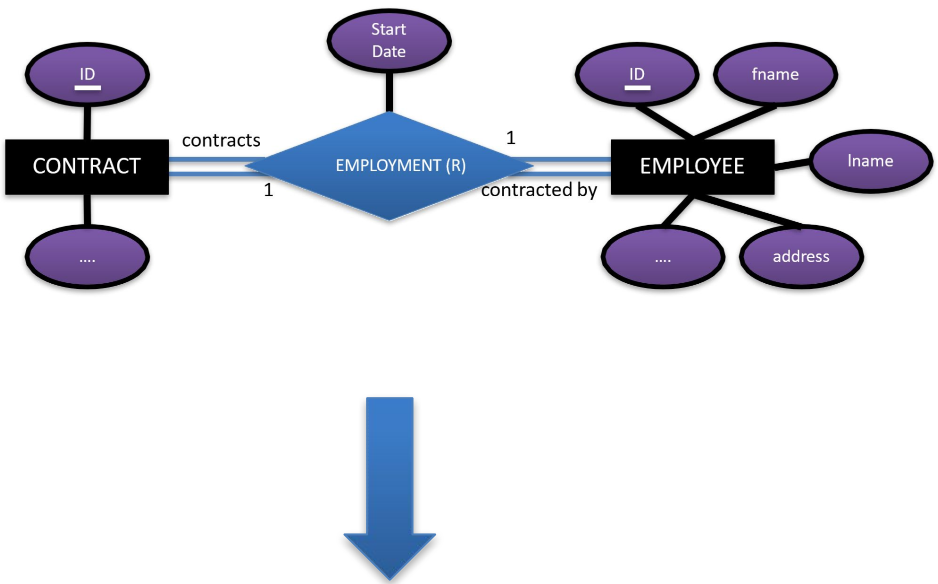


- Foreign Key Approach
  - choose one of the participating relations, say S
  - include as a foreign key in S the primary key of T
  - if possible, choose an entity type with *total participation* in R for the role of S
  - include all the simple attributes of the relationship type R as attributes of S

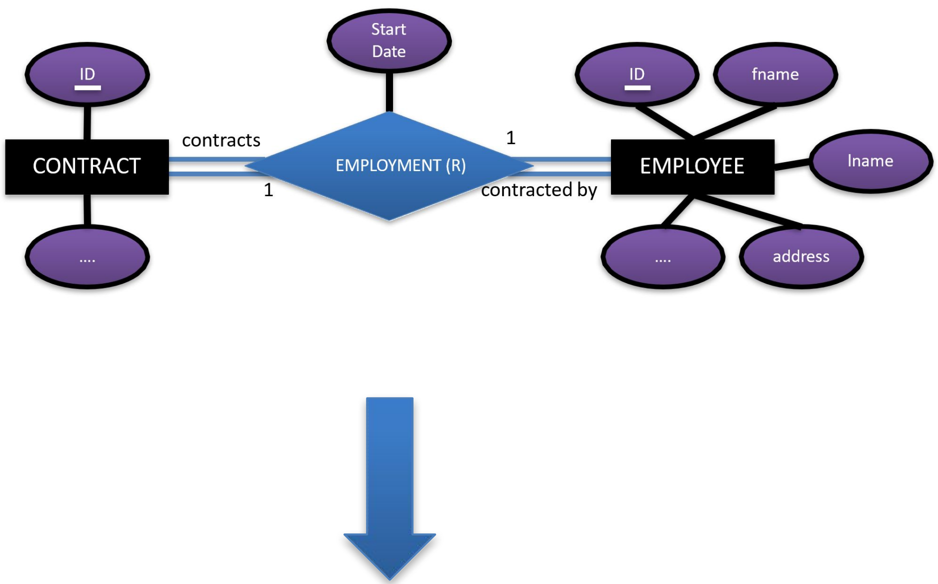
CONTRACT (T)

contract_id	….

EMPLOYEE (S)

employee_id	fname	lname	address	….	contract_id	start date

Mapping 1:1 Relationships



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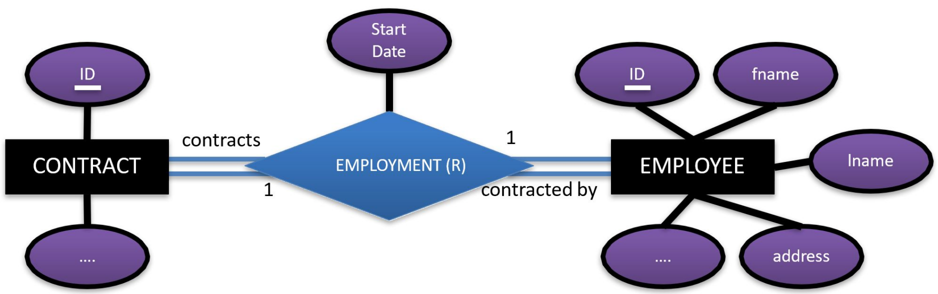
CONTRACT (T)

<u>contract_id</u>	....

EMPLOYEE (S)

<u>employee_id</u>	fname	lname	address	....		

Mapping 1:1 Relationships



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EMPLOYEE (S)

employee_id	fname	lname	address	....		

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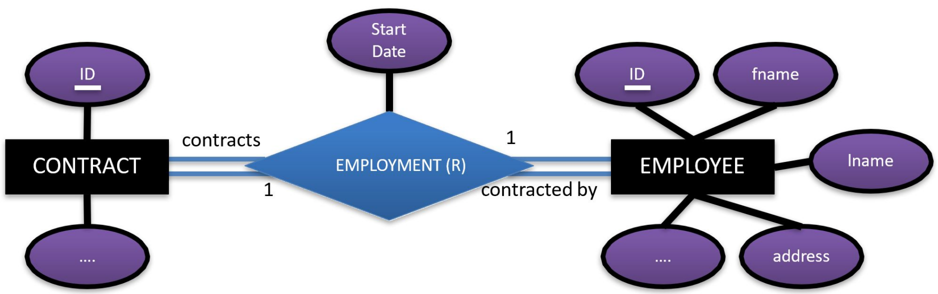

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<u>contract_id</u>	....

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<u>employee_id</u>	fname	lname	address	....		

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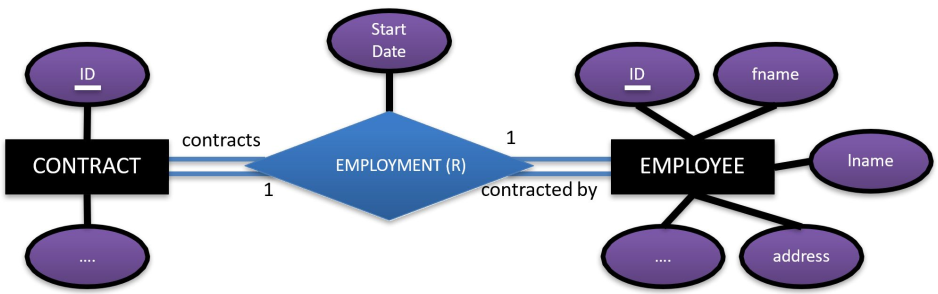

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contract_id	....

EMPLOYEE (S)

employee_id	fname	lname	address	....	contract_id	

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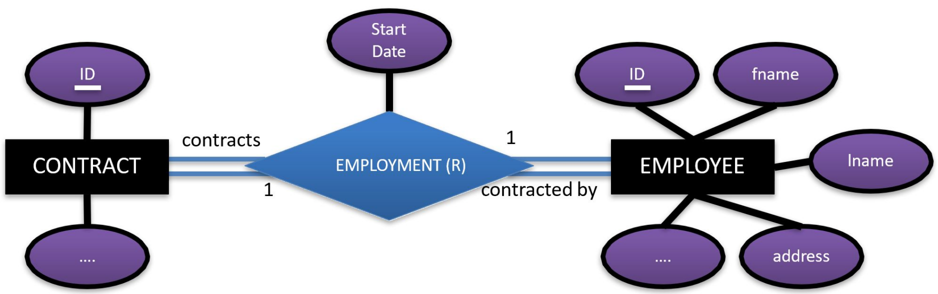

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CONTRACT (T)

<u>contract_id</u>	....

EMPLOYEE (S)

<u>employee_id</u>	fname	lname	address	....	contract_id	

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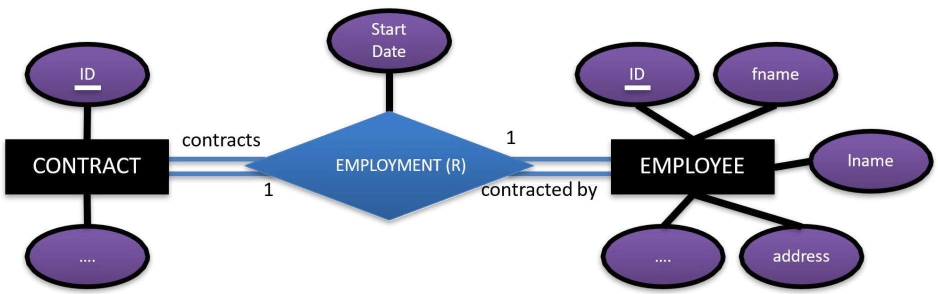

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employee_id	fname	lname	address	….	contract_id	

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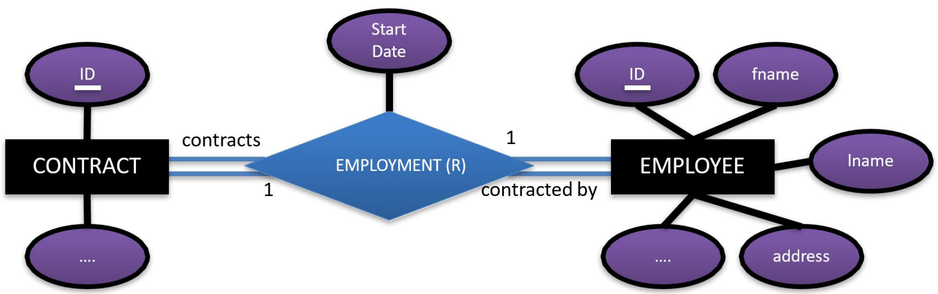

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EMPLOYEE (S)

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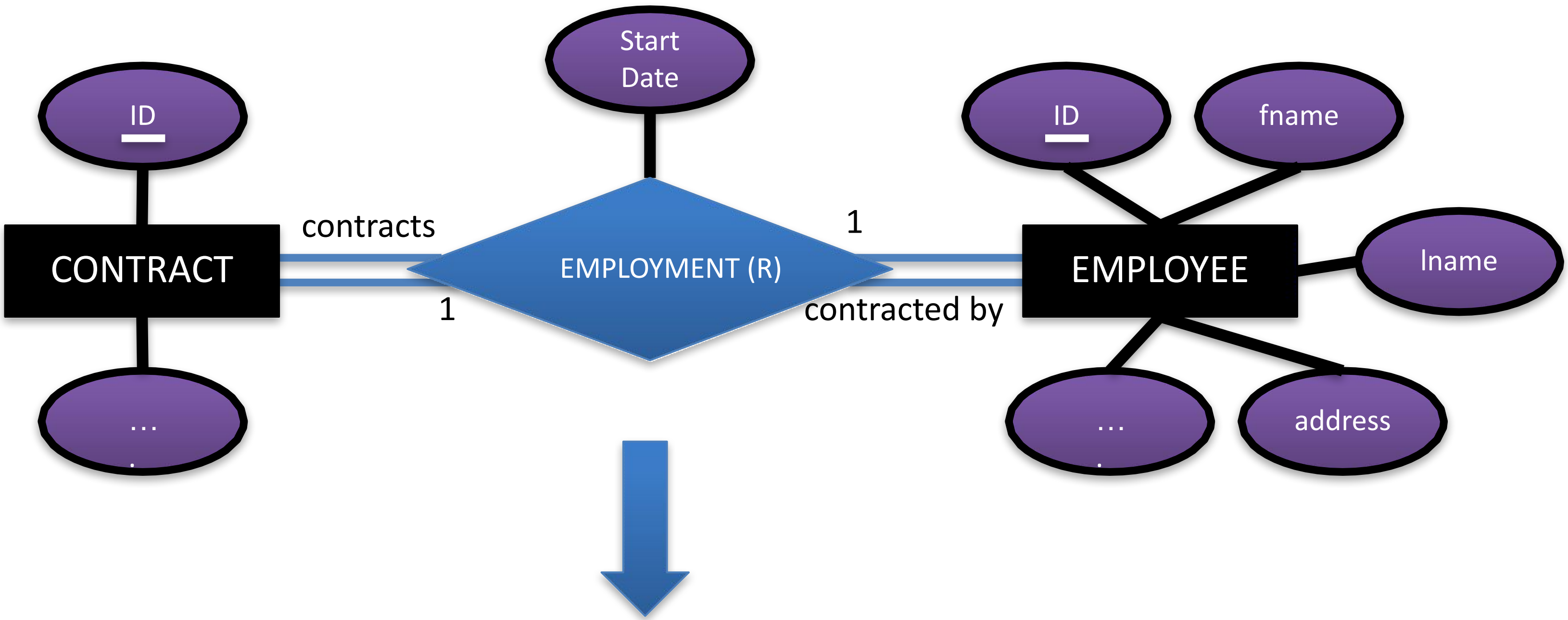

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Mapping 1:1 Relationships

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  - This can only be used when both S and T have *total participation* in the relationship type R
  - Merge the two entity types S and T and the relationship type R into one single relation V
  - V should include all the simple component attributes of S, T and R
  - This is possible as the joint total participation indicates that the two tables will have an identical number of tuples at all time

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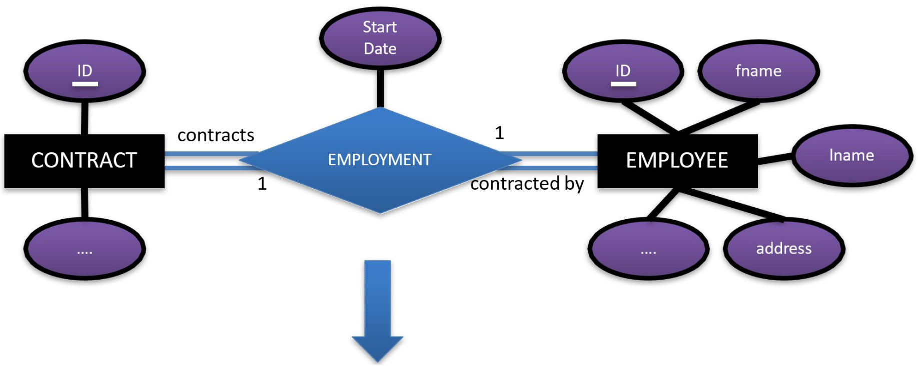

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EMPLOYEE_RECORD

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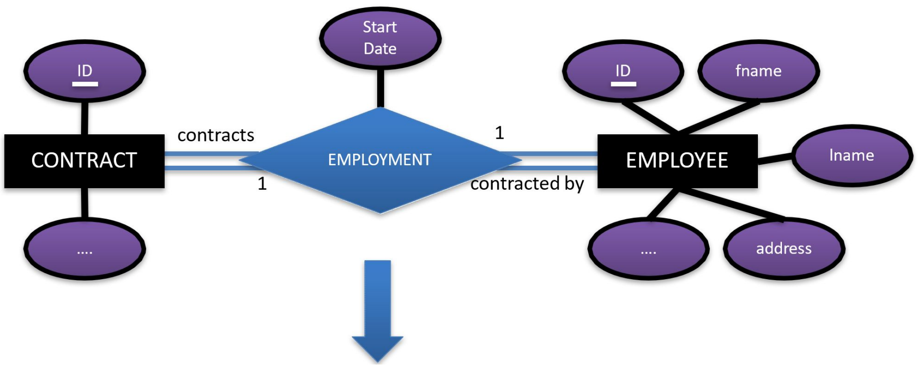

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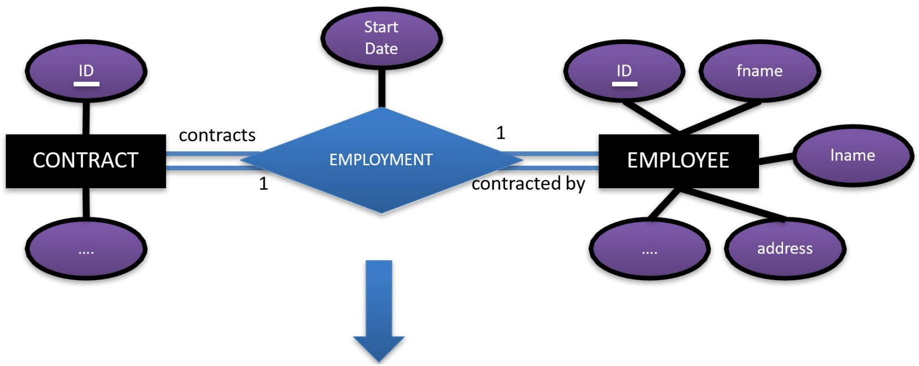

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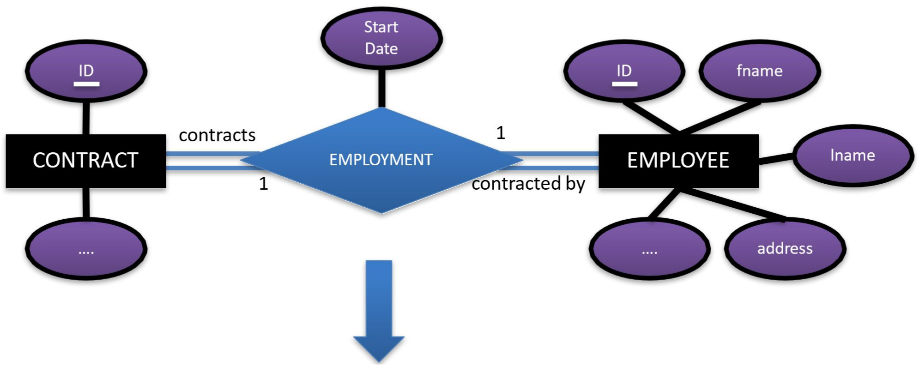

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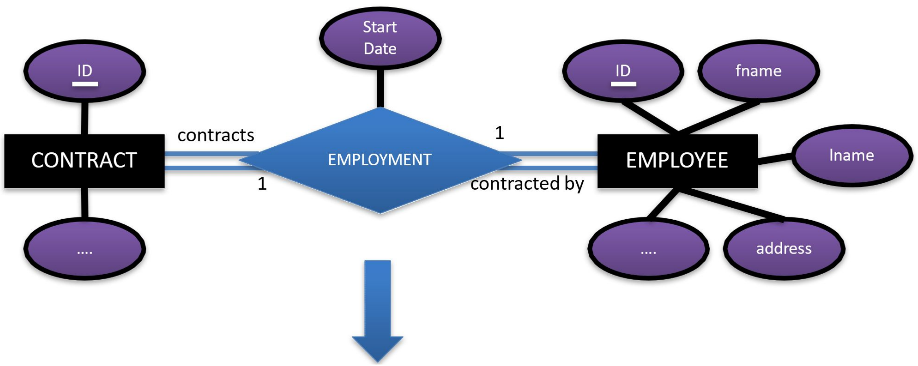

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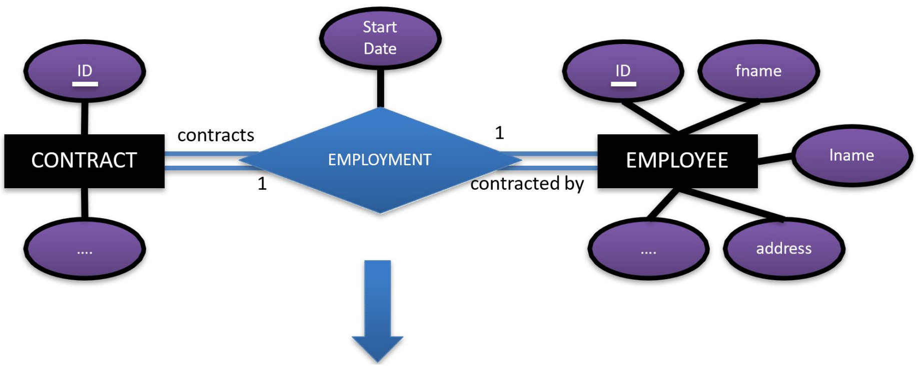

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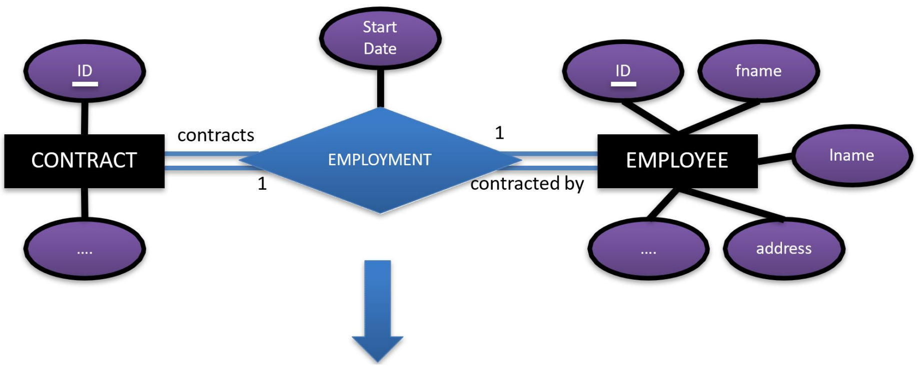

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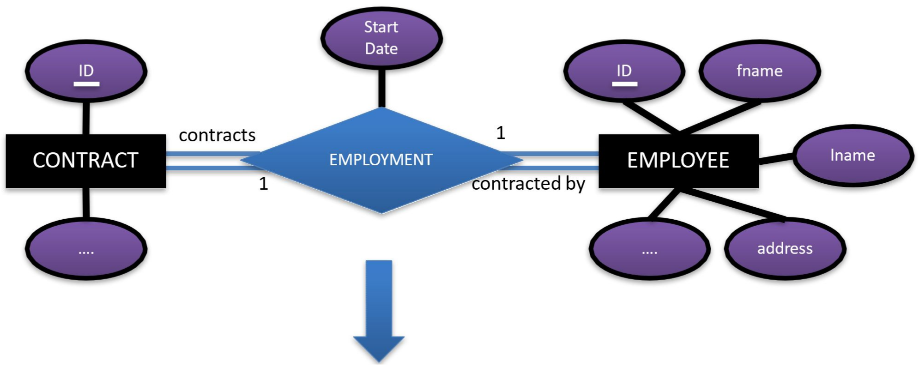
EMPLOYEE_RECORD

contract_id	….	start_date	employee_id	fname	lname	address	….

S

Mapping 1:1 Relationships



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EMPLOYEE_RECORD

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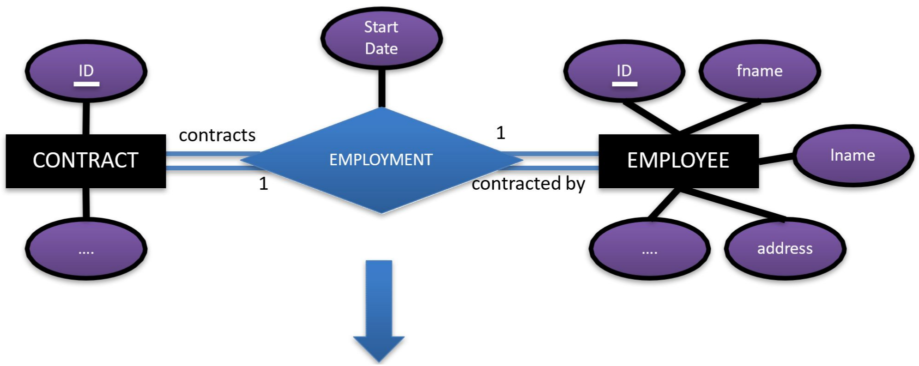
T

S

Mapping 1:1 Relationships



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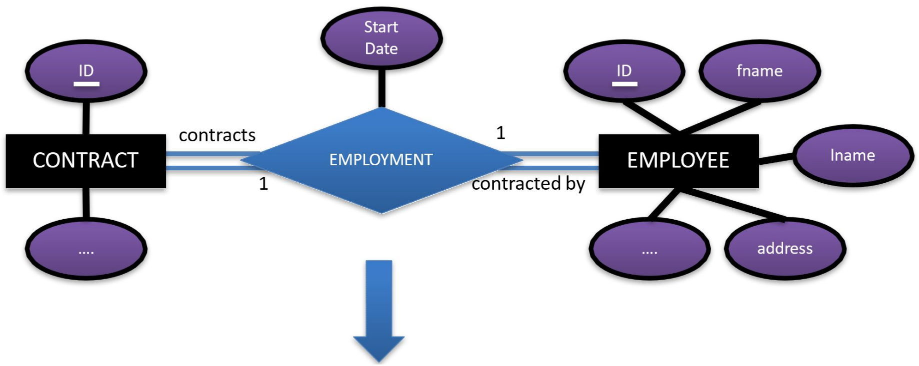
T

R

S

Mapping 1:1 Relationships



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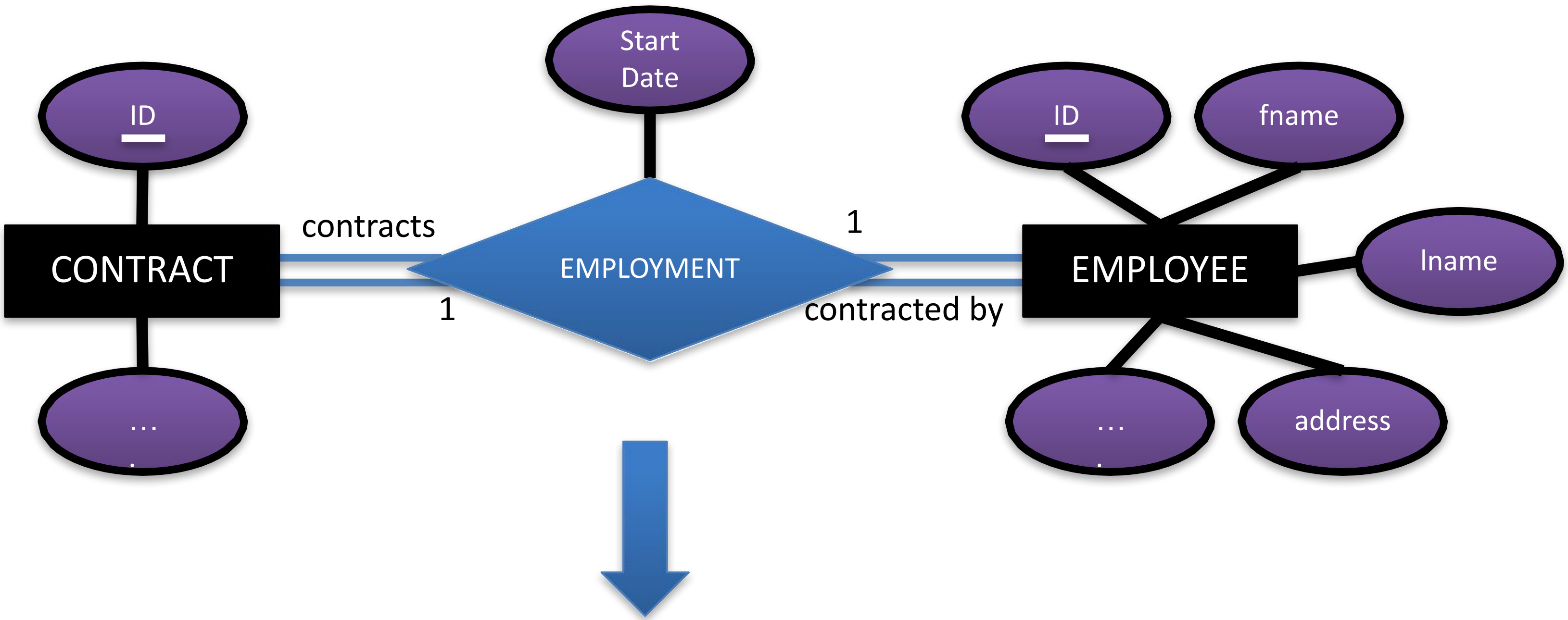

EMPLOYEE_RECORD

contract_id	….	start_date	employee_id	fname	lname	address	….

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Mapping 1:N Relationships

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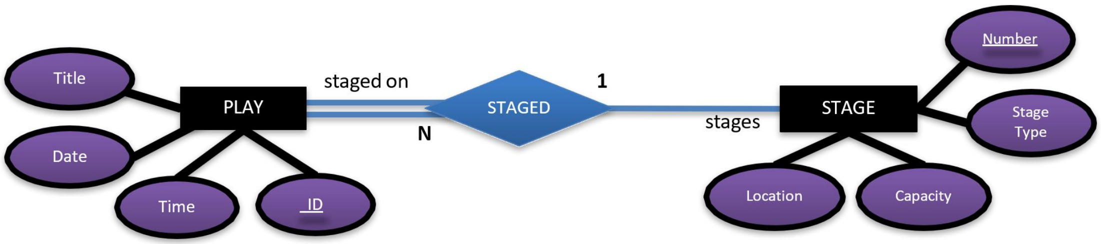

- For each binary 1:N relationship type R
  - identify the relation S that corresponds to the entity types on the N-side of R
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- Include any simple attributes of the relationship type R as attributes of S
  - or simple component attributes of a composite attribute

PLA
Y

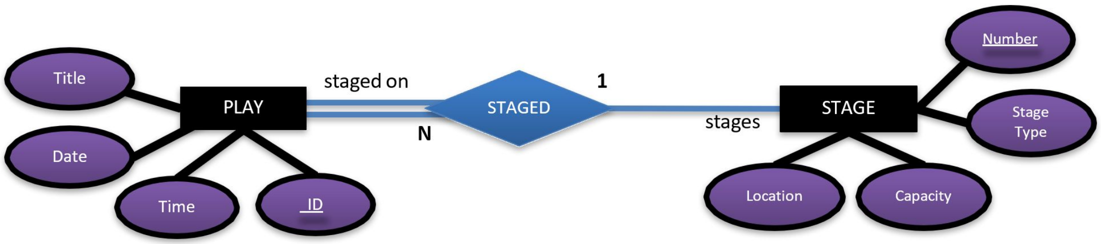
play_id	time	date	stage_number

STAGE

number	stage_type	capacity	location

51

Mapping 1:N Relationships



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PLA
Y

play_id	time	date	

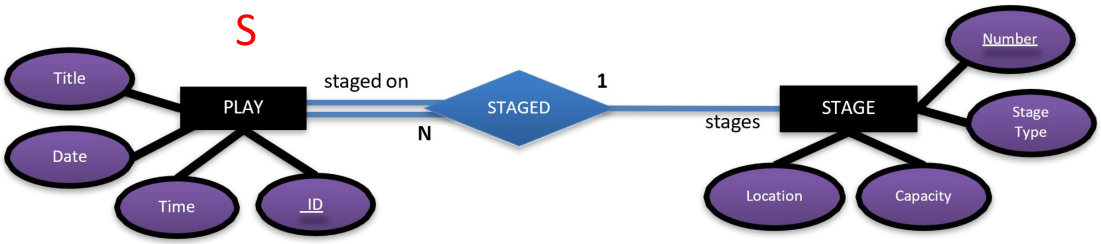
STAGE

number	stage_type	capacity	location

52

Mapping 1:N Relationships

S



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S

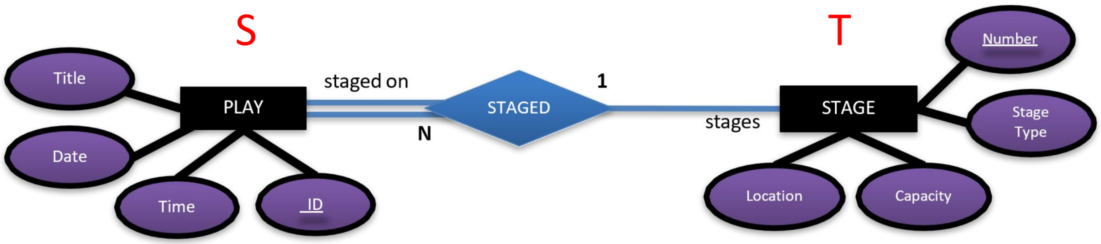
PLAY

play_id	time	date	

STAGE

number	stage_type	capacity	location

Mapping 1:N Relationships



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S

PLAY

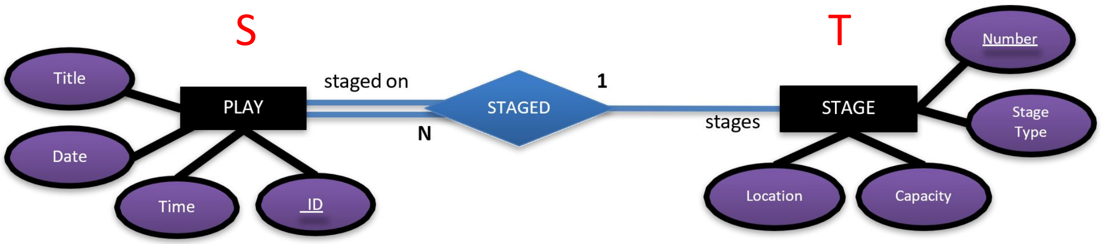
play_id	time	date	

T

STAGE

number	stage_type	capacity	location

Mapping 1:N Relationships



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PLAY

play_id	time	date	

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STAGE

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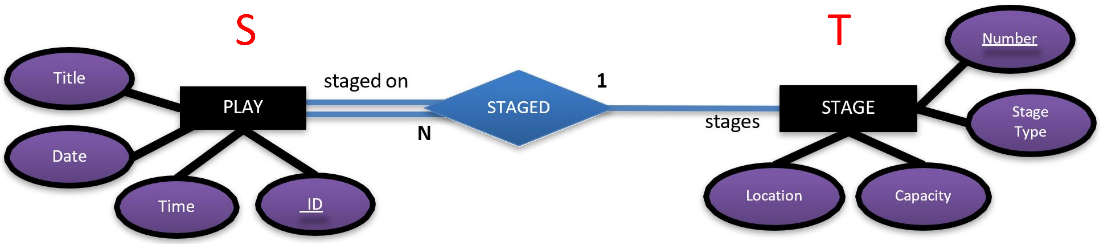
Mapping 1:N Relationships



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PLAY

play_id	time	date	stage_number

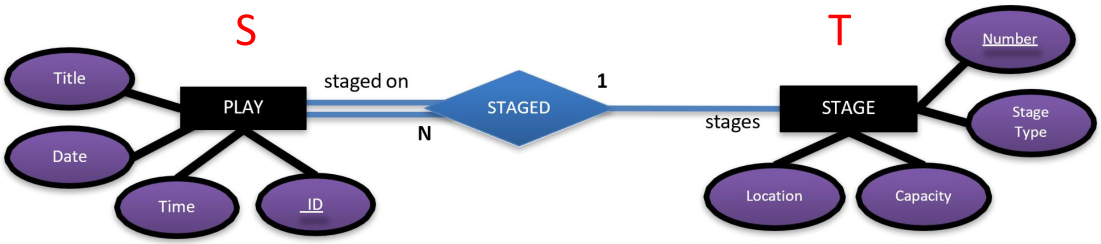
T

STAGE

number	stage_type	capacity	location

56

Mapping 1:N Relationships



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S

PLAY

play_id	time	date	stage_number

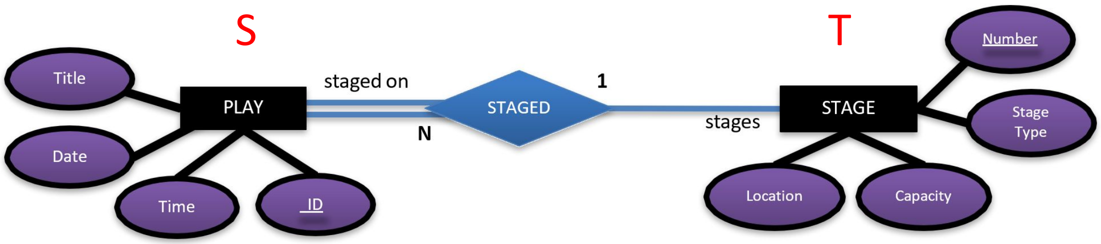
T

STAGE

number	stage_type	capacity	location

57

Mapping 1:N Relationships



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S

PLAY

play_id	time	date	stage_number

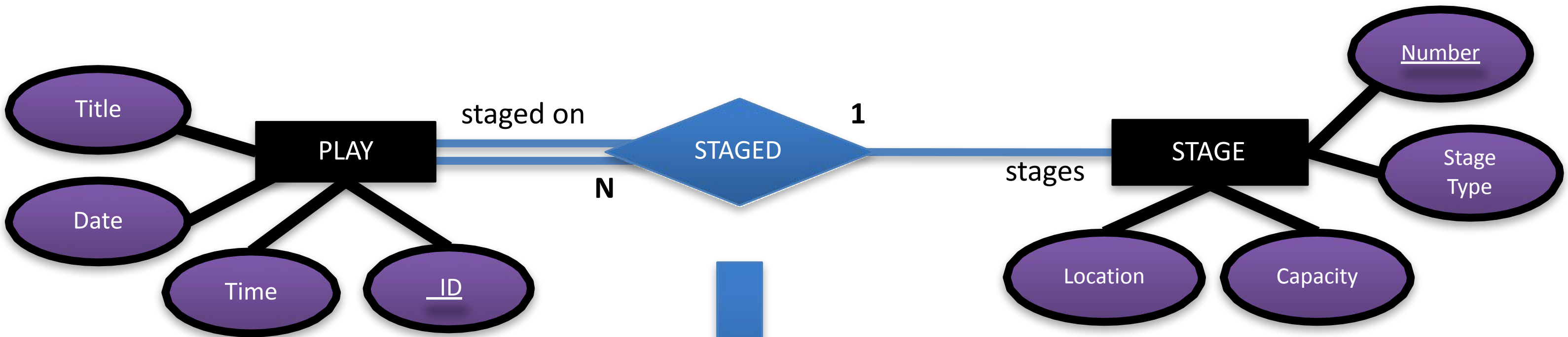
T

STAGE

number	stage_type	capacity	location

58

Mapping 1:N Relationships



PLAY (S)

play_id	title	time	date

STAGE (T)

number	stage_type	capacity	location

PLA
Y

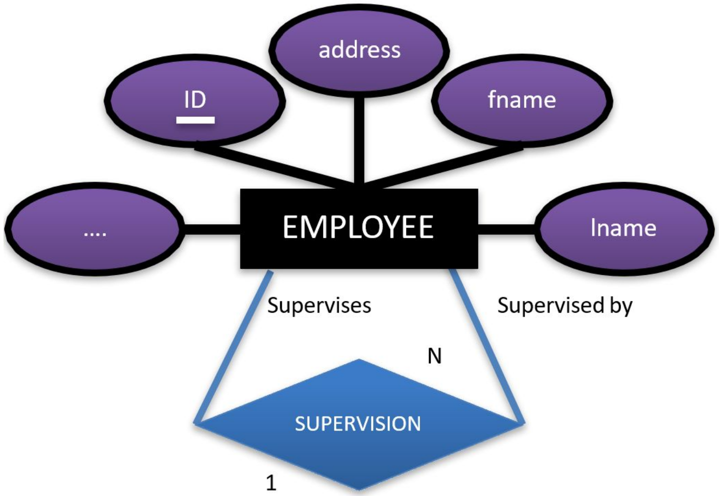
play_id	time	date	stage_number

STAGE

number	stage_type	capacity	location

Mapping Recursive Relationships

Mapping Recursive Relationships



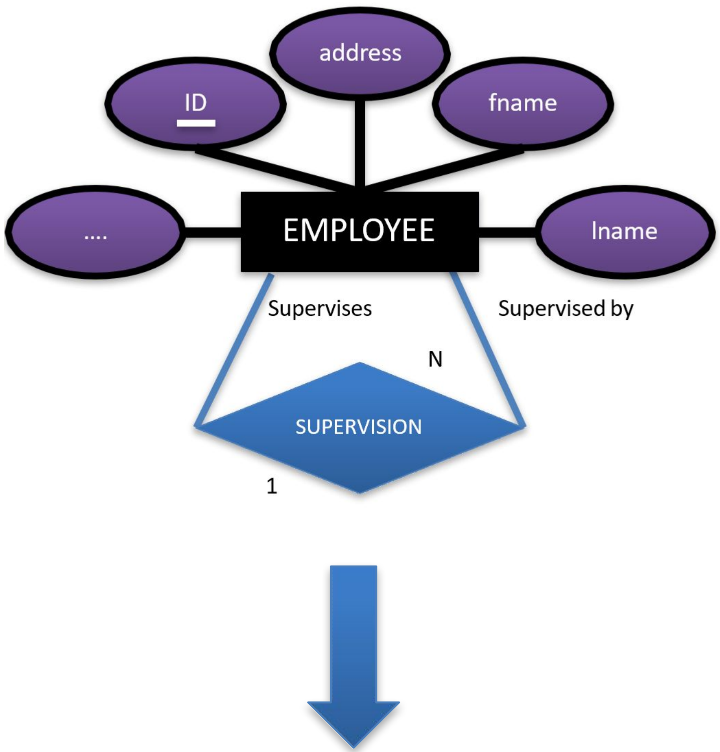
- Recursive relationships
  - where an entity instance can refer to another instance of the same entity type
- For each recursive relationship type R
  - Identify T, the Entity type
  - Include the primary key of T as a foreign key in the same relation, T
  - Include any simple attributes of the relationship type
    R as attributes of T
    - or simple component attributes of a composite attribute

EMPLOYEE (T)

employee_id	fname	lname	address	supervisor_id	....

61

Mapping Recursive Relationships



EMPLOYEE

employee_id	fname	lname	address	....

- Recursive relationships
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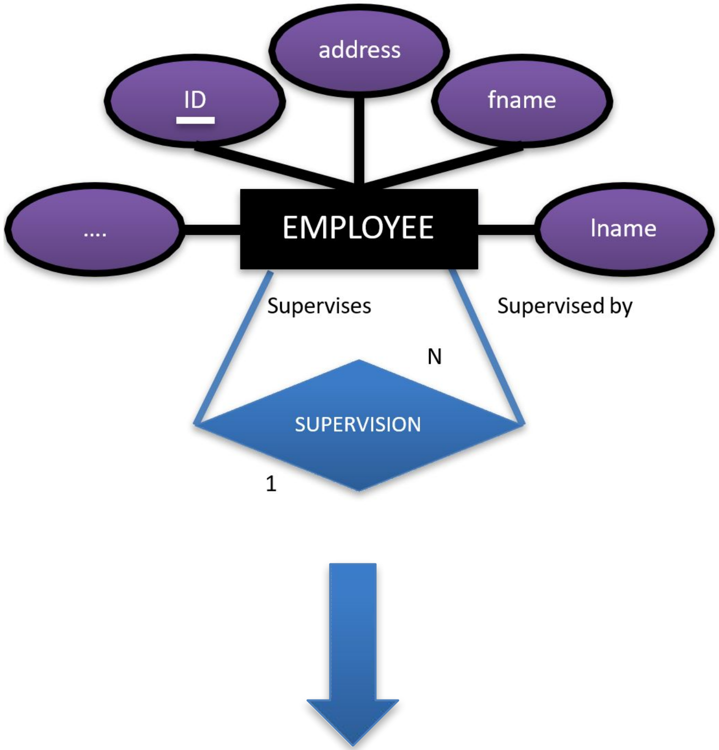
62

Mapping Recursive Relationships



EMPLOYEE (T)

employee_id	fname	lname	address	....

- Recursive relationships
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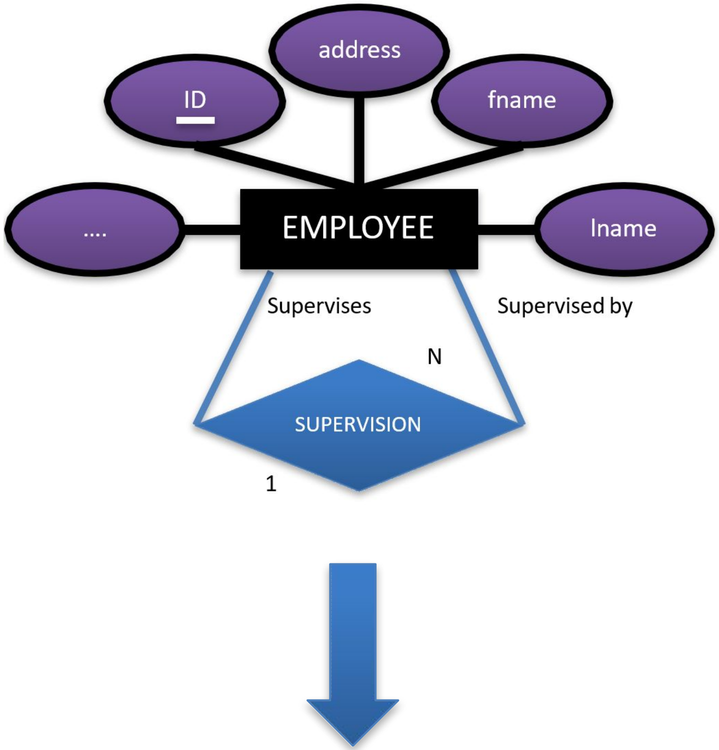
63

Mapping Recursive Relationships



EMPLOYEE (T)

employee_id	fname	lname	address	....

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  - where an entity instance can refer to another instance of the same entity type
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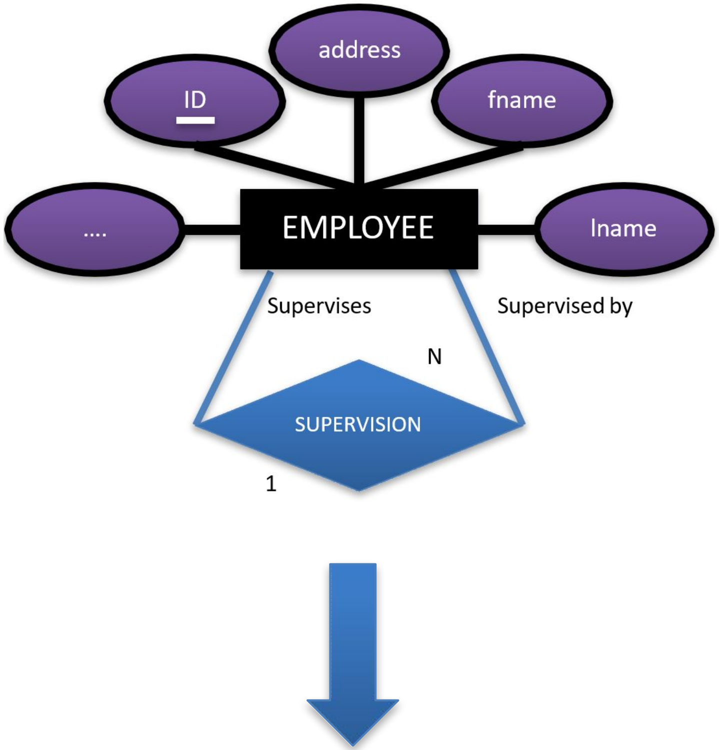
Mapping Recursive Relationships



EMPLOYEE (T)

employee_id	fname	lname	address	employee_id	….

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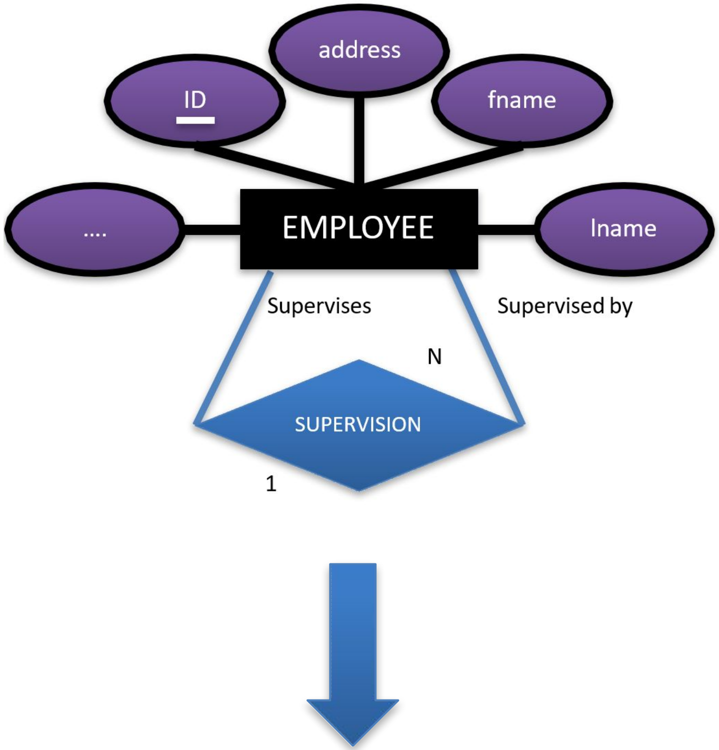

EMPLOYEE (T)

employee_id	fname	lname	address	supervisor_id	....

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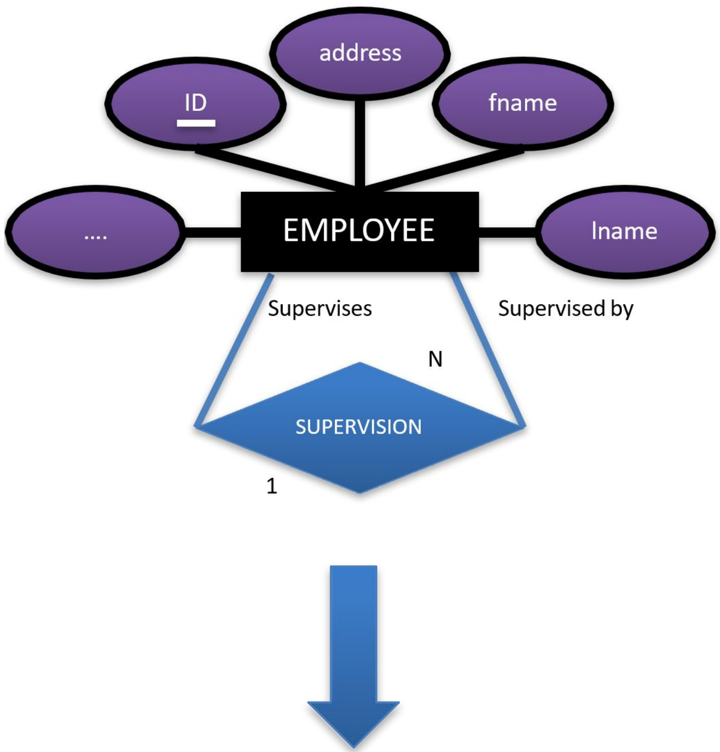
Mapping Recursive Relationships



EMPLOYEE (T)

employee_id	fname	lname	address	supervisor_id	....

- Recursive relationships
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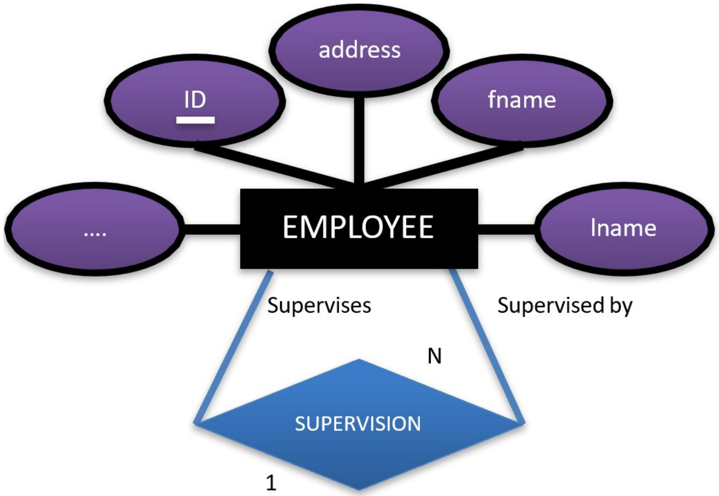
Mapping Recursive Relationships



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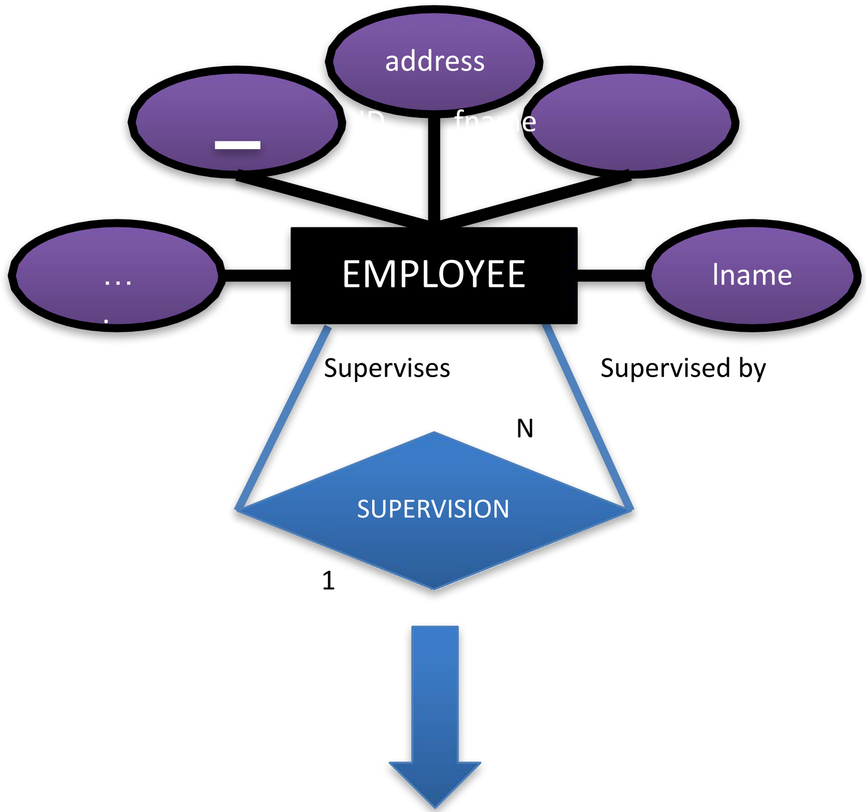
EMPLOYEE (T)

employee_id	fname	lname	address	supervisor_id	....

68

Mapping Recursive Relationships



EMPLOYEE (T)

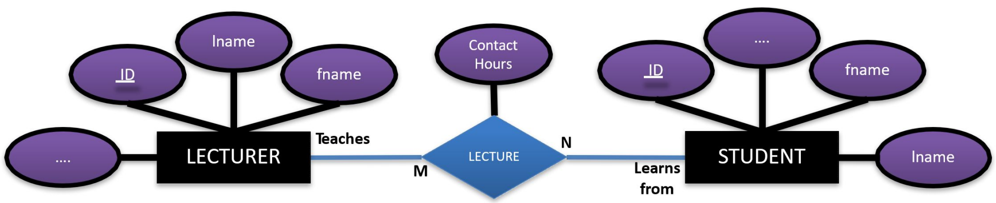
employee_id	fname	lname	address	supervisor_id	….

Mapping M:N Relationships

Mapping M:N Relationships

- Many to Many relationship types are more complex to map than 1:1 or 1:N
- As each entity instance may reference many entity instances in the other participating entity type
  - You cannot use a foreign key attribute in either participating entity
  - You must create a new relation to represent the relationship type

Mapping M:N Relationships



- For each binary M:N relationship type R
  - create a new relation S to represent R
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types
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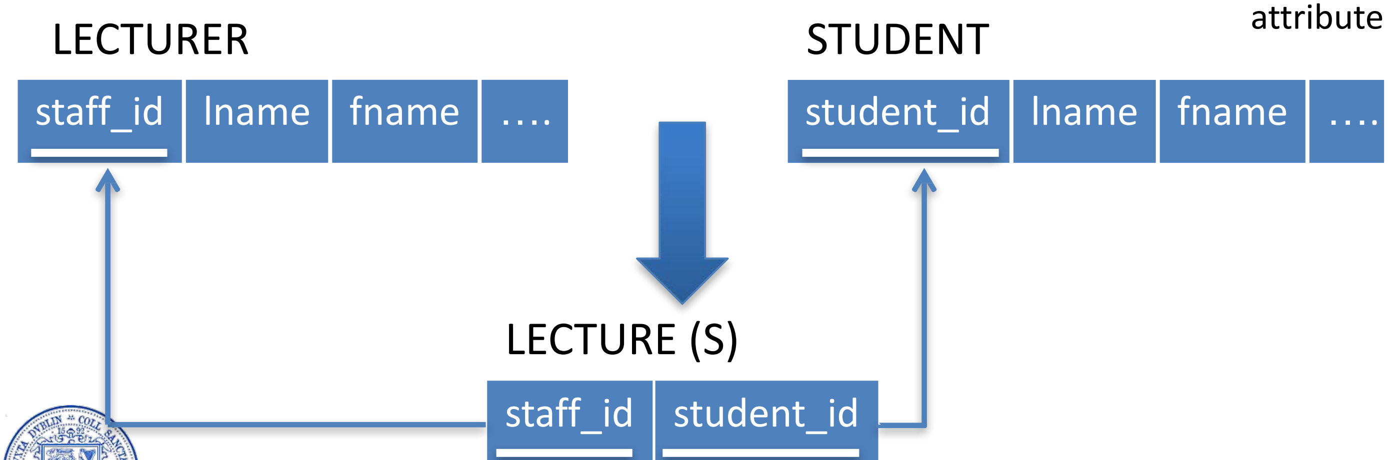

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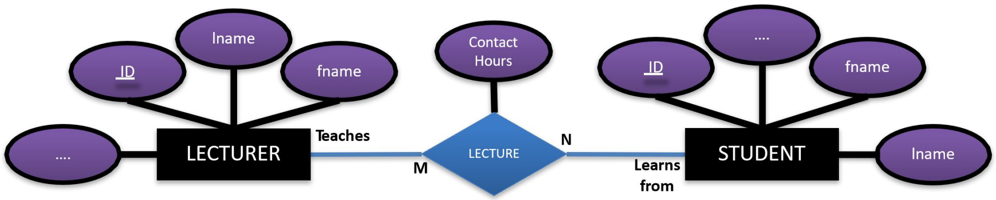

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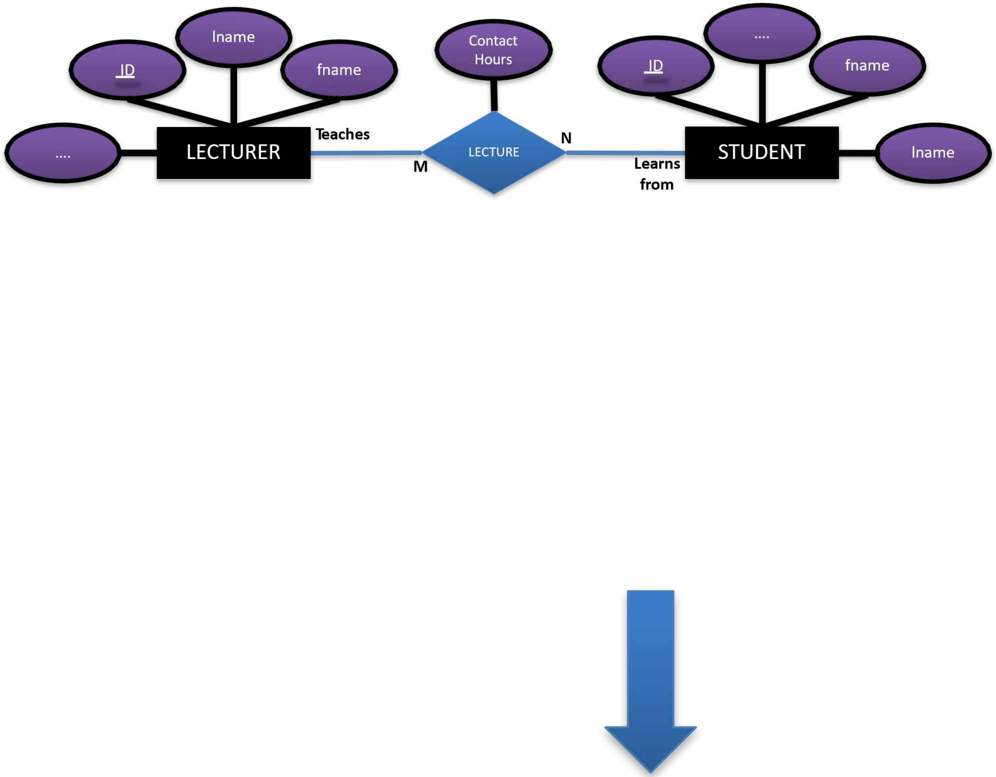

LECTURE (S)

- For each binary M:N relationship type R
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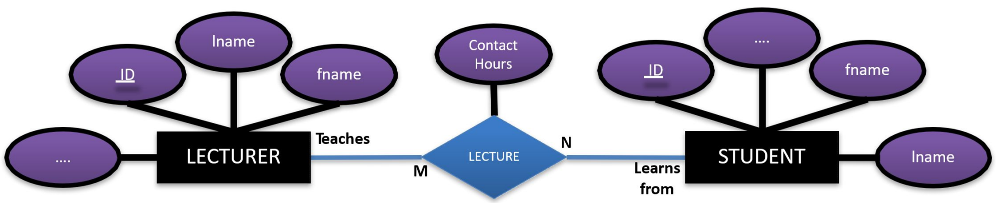

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LECTURER

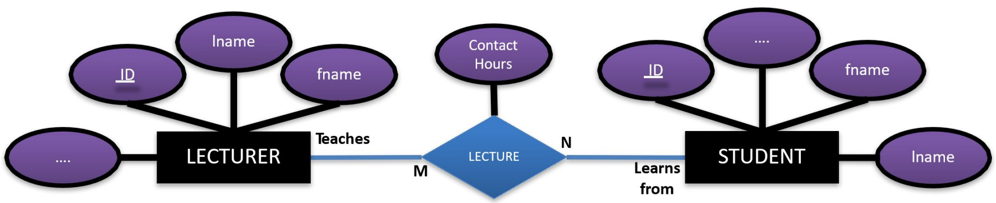
staff_id	lname	fname	....

STUDENT

student_id	lname	fname	....

LECTURE (S)

Mapping M:N Relationships



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LECTURER

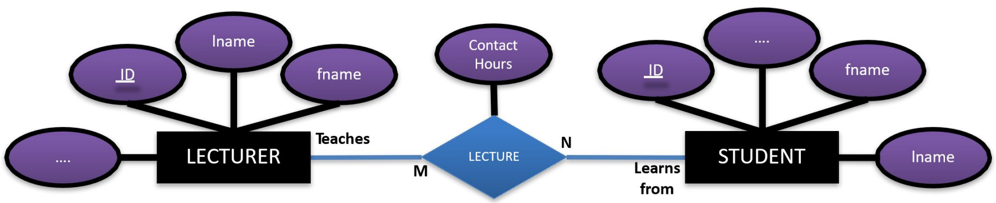
staff_id	lname	fname	....

STUDENT

student_id	lname	fname	....

LECTURE (S)

Mapping M:N Relationships



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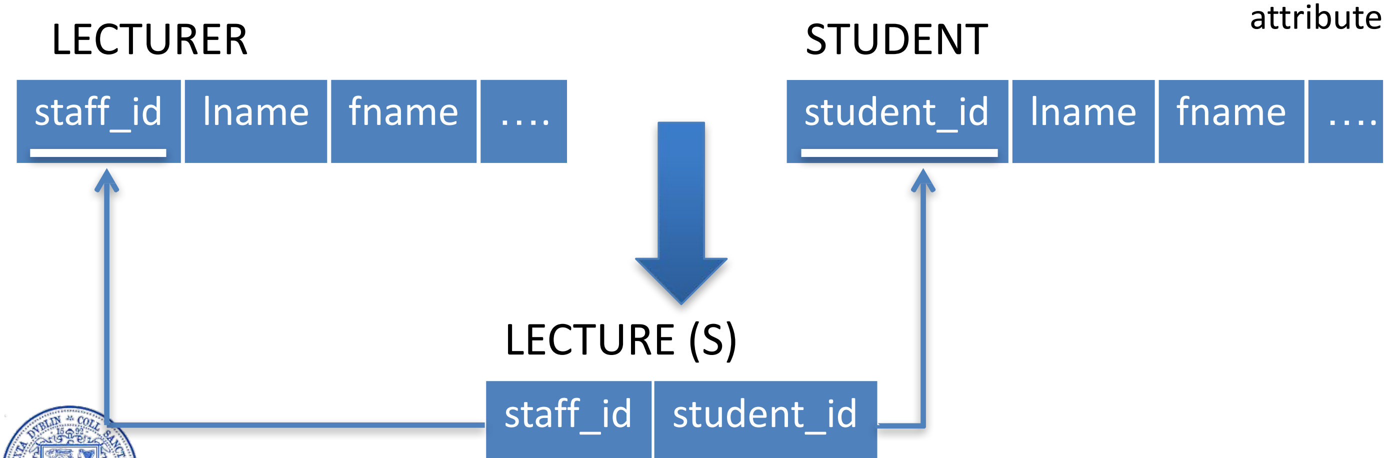

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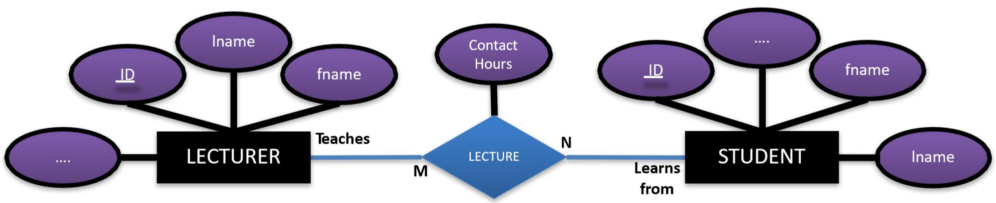

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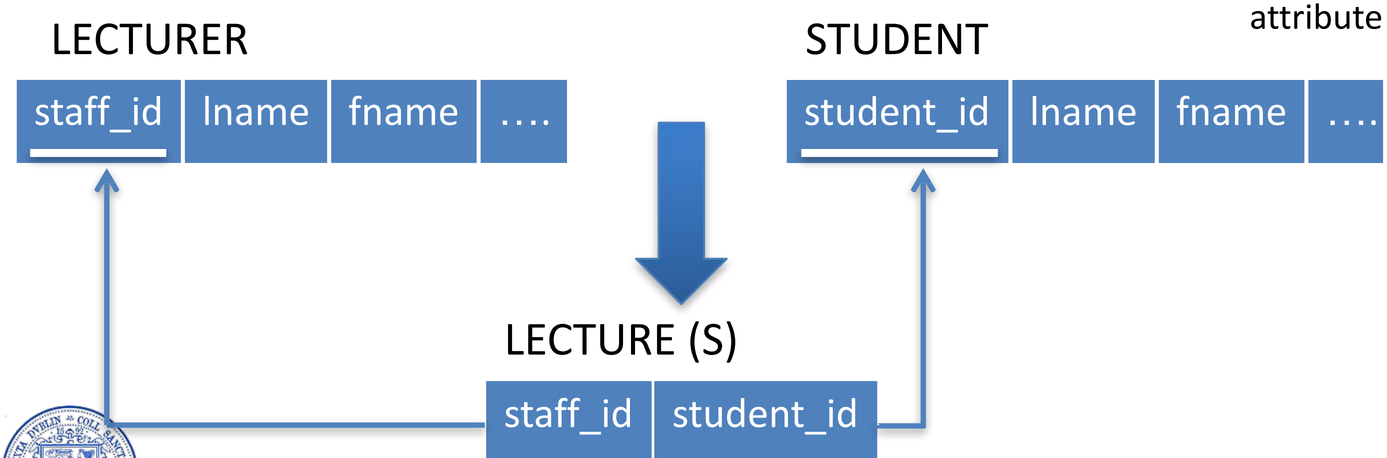

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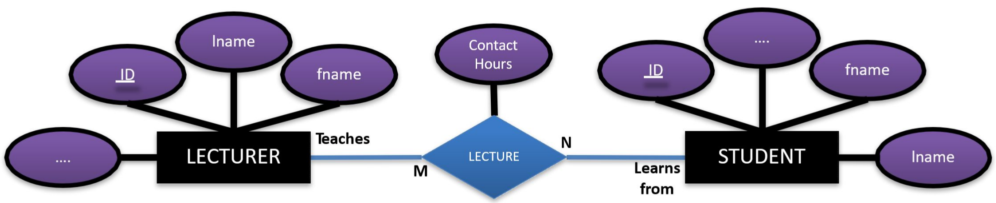

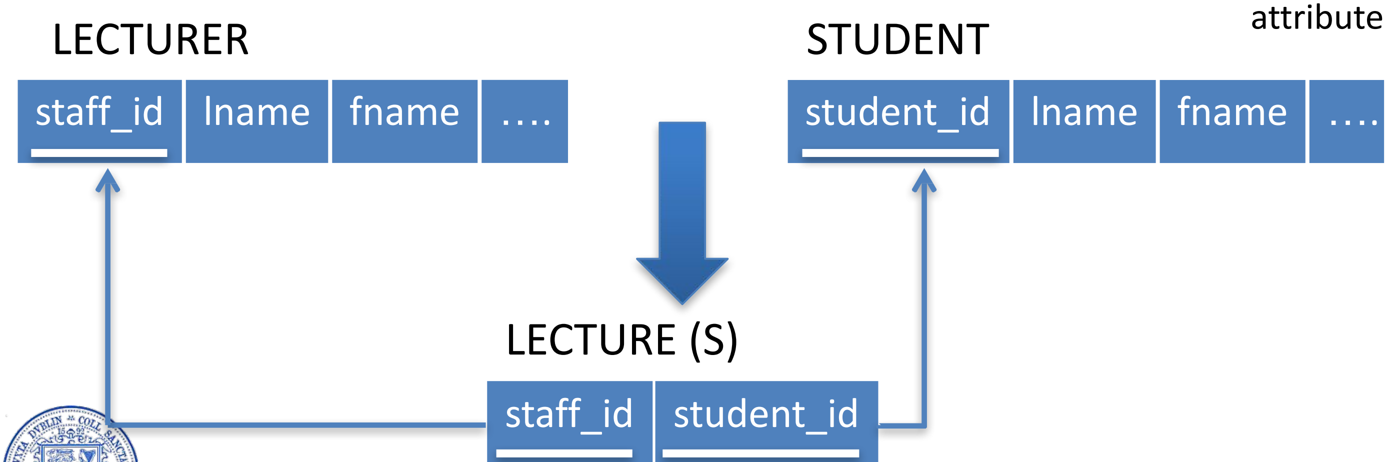

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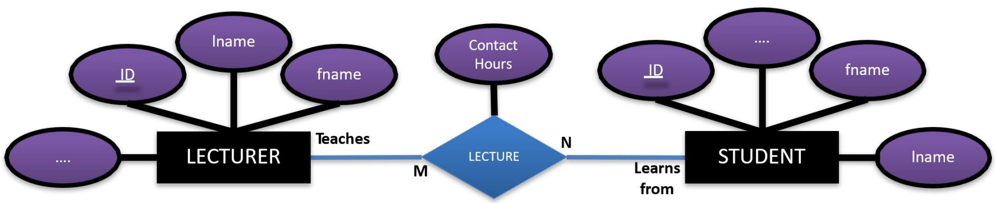

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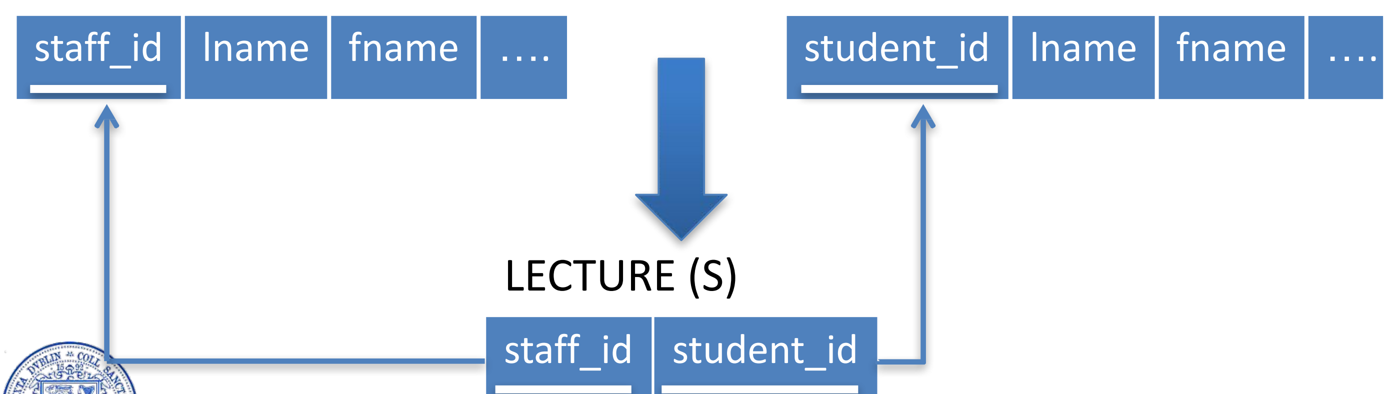
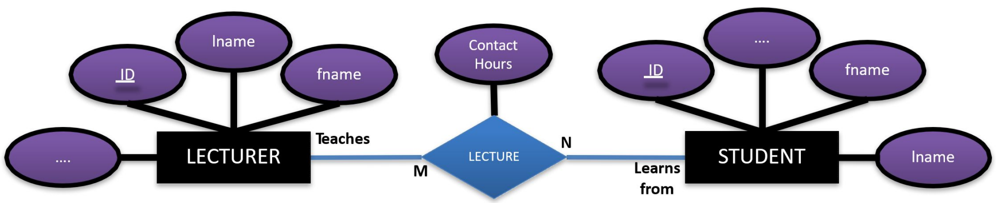
LECTURER

STUDENT



Mapping M:N Relationships



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LECTURER

staff_id	lname	fname	....

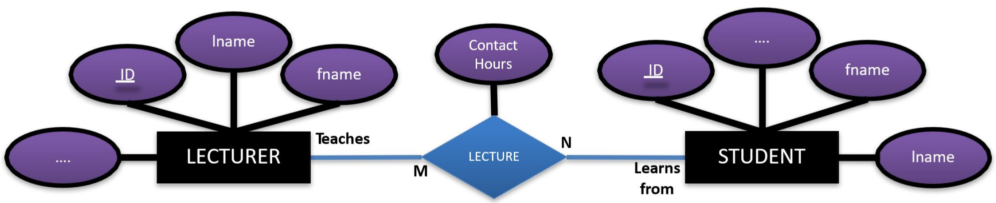
STUDENT

student_id	lname	fname	....

LECTURE

staff_id	student_id	contact_hours

Mapping M:N Relationships



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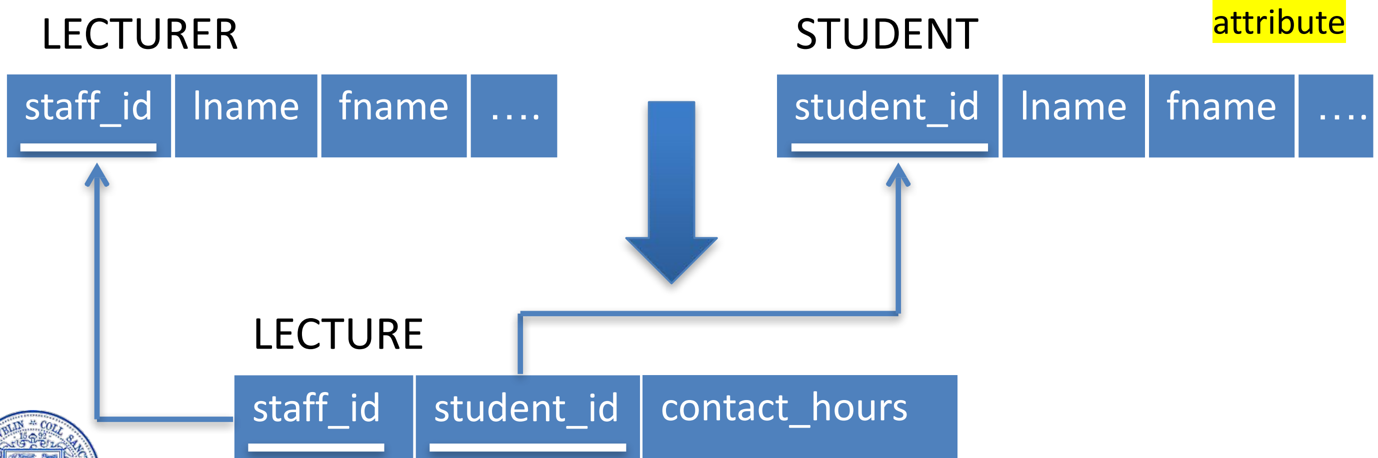

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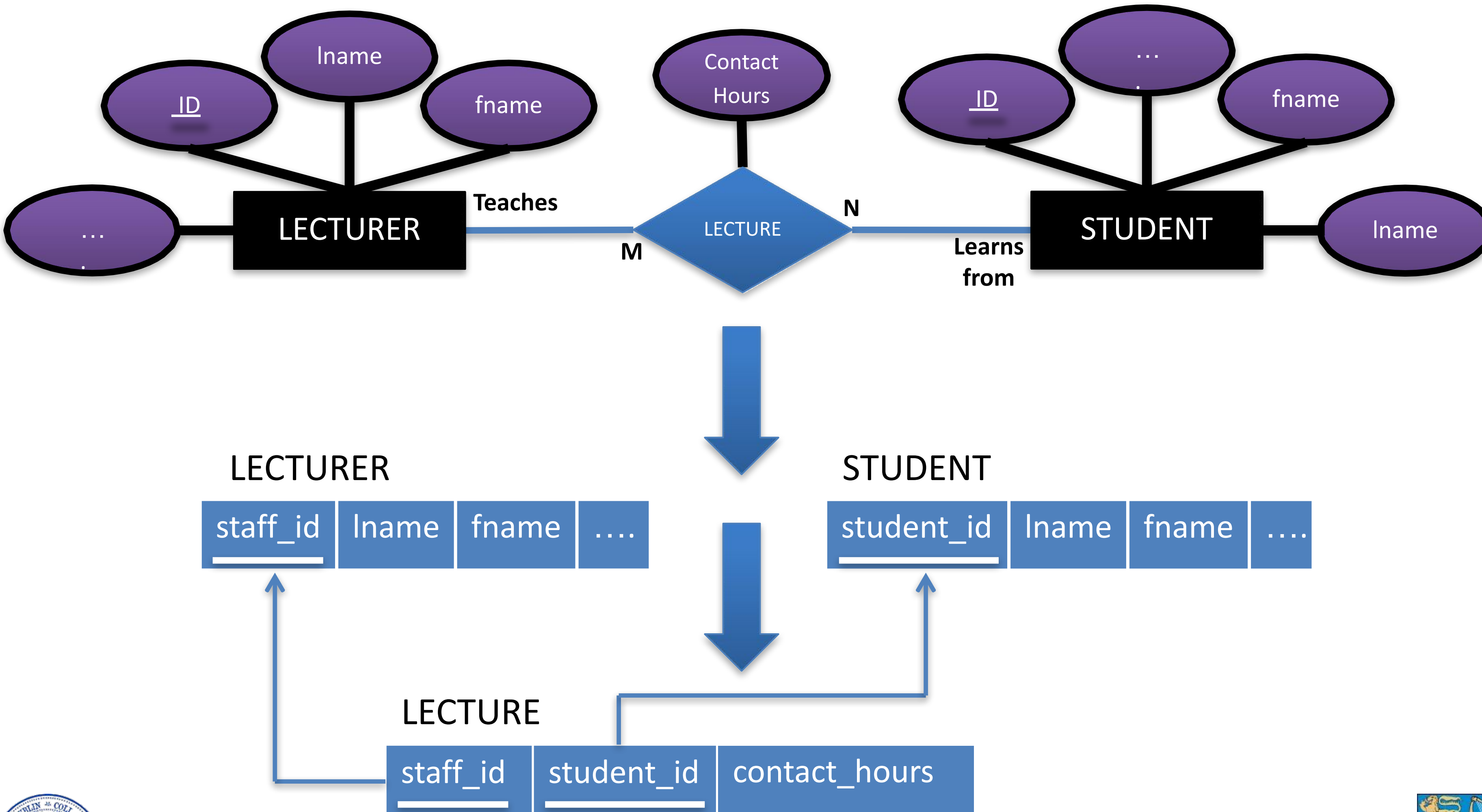
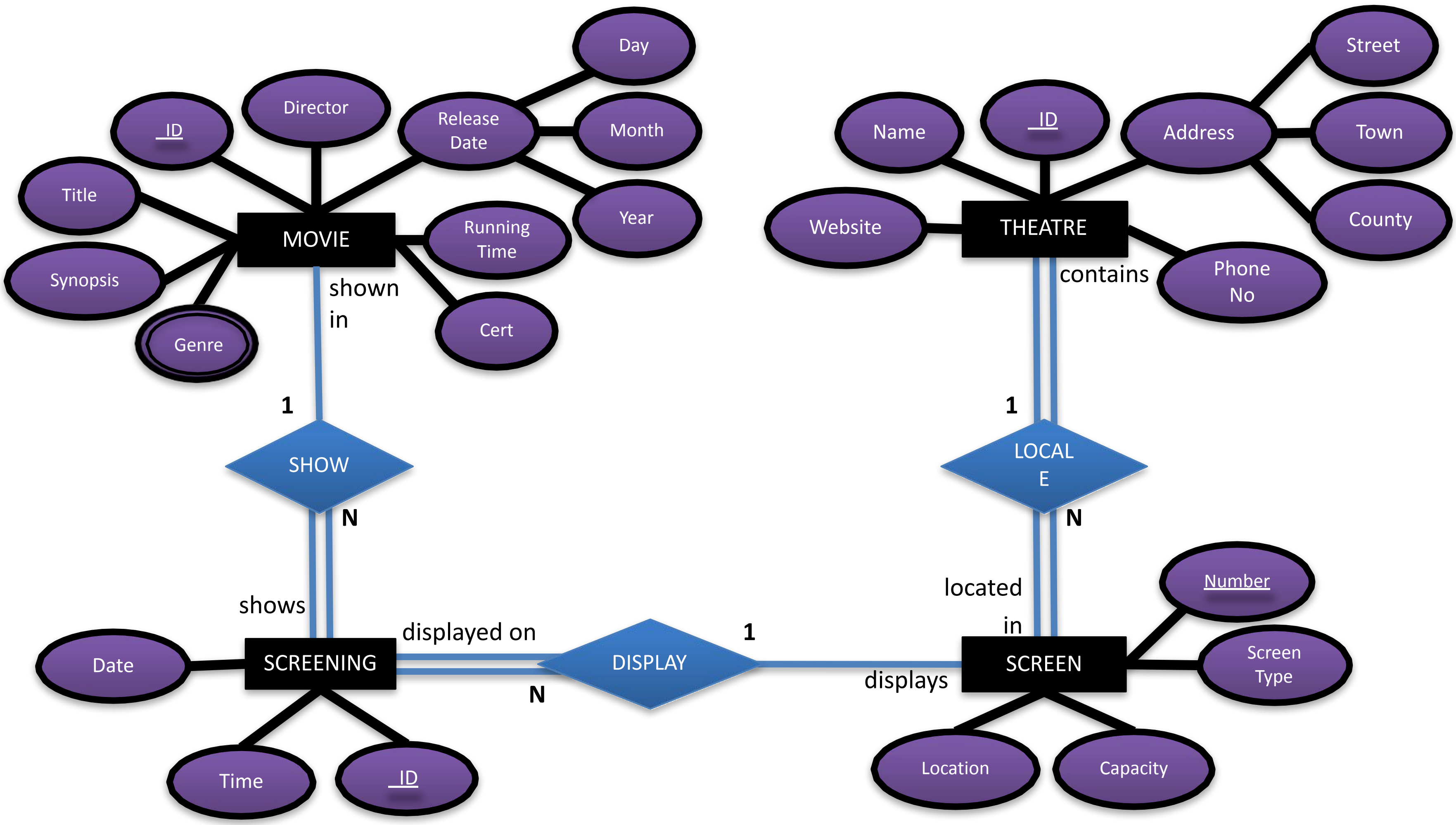


Summary

- Each entity – create a table
- Composite attributes – include the simple ones as an attribute (don’t combine)
- Multivalued attribute – create additional table with FK and composite primary key
- 1 to 1 relationships
  - FK approach – no extra table – include FK of one entity in other
  - Merged Relation approach (total participation of both needed) – merge the 2 tables
- 1 to N relationships
  - Include in the N-side table FK; include simple attributes in N-side table
- Many to many
  - Create additional table; two FKs in new table form composite PK

Cinema Example

Cinema Example: Take some time to come up with the correct tables given the mapping rules



Cinema Example